

Medical Robotics

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Lecture 4

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1. Introduction: From Surgical Tasks to Kinematic Architectures

The design of a medical robot begins not with the hardware, but with the clinical task. In previous lectures we constructed a reference table mapping *domains of use* to *robotic tasks*, *kinematic architectures*, *control modalities*, and *sensors and actuators*. This lecture focuses on the third column of that table — the kinematic architecture — and asks: given a surgical task with its associated constraints, which manipulator structure best satisfies them?

The question is answered by a combination of empirical survey (looking at what has been built and deployed) and principled analysis (understanding why a given architecture fits or fails to fit a given task). Both approaches are presented here.

Why architecture matters. Surgical robots are not reprogrammable industrial arms placed in an operating room. They must satisfy constraints that industrial robots never face: workspace shape fitted to the human body, intrinsic safety in case of power failure, compatibility with imaging systems, singularity-free operation within the intervention volume, and inverse kinematics solvable in closed form for speed and reliability. Each of these constraints influences the kinematic architecture at the design stage.

2. Survey of Kinematic Architectures in Medical Robotics

2.1. Domains of Use and Prevalence

A survey of approximately 100 medical robots reveals that **minimally invasive surgery (MIS)** accounts for the largest share of application domains — around 46% — followed by neurosurgery (21%), orthopedics (12%), ultrasound/echography (10%), needle/biopsy (8%), and radiotherapy (3%). This distribution reflects a key historical insight: in MIS, robots genuinely enabled something previously impossible — restoring the dexterity of open surgery while maintaining the minimal access that laparoscopy introduced. In other domains such as orthopedics, the contribution of robots was more akin to the accuracy improvements already familiar from industrial applications.

Regarding kinematic structure, the survey shows that **serial** architectures dominate at 61%, followed by **hybrid** (parallel + serial) at 24%, and **parallel** architectures at 15%. Serial architectures remain prevalent across medical robotics, just as they do in industrial applications. Parallel architectures, while less common in standard robotics courses, are particularly relevant in the medical domain and will be treated in dedicated sessions of this course.

2.2. The Reference Design Table

The full mapping of surgical domains to kinematic specifications is summarised in Table 1. This table is the principal reference framework of the course.

3. Conventional Serial Architectures: SCARA vs. Anthropomorphic

3.1. The Two Main Types

Two conventional serial architectures recur most frequently in medical robotics: the **SCARA** (Selective Compliance Assembly Robot Arm) and the **anthropomorphic** (elbow-type) manipulator. Both typically have three degrees of freedom (DOF) for positioning and two or three additional DOF for the wrist, yielding five or six DOF in total.

Representative systems:

Domain	Robotic task	Kinematic architecture	Control modality	Sensors & actuators
Orthopedics	Machining of rigid surface	Conventional or parallel + dedicated wrist; 5 (drill) or 6 (cut) DOF	Autonomous, cooperative, or “hands-on”	Conventional; vel: mm/s; force: up to 100 N
MIS	Constrained manipulation	RCM; passive joints; 5 or 6 external + extra internal DOF	Teleoperation, shared control	Conventional; vel: cm/s; force: few N
Neurosurgery, interventional radiology, radiotherapy	Constrained targeting	Conventional + front-end with dedicated architecture; 5 or 6 DOF	Teleoperation, (semi-)autonomous	MR/CT compatible (pneumatic, ultrasonic); force: few N; insertion usually manual
Microsurgery	Micromanipulation	Dedicated kinematic architecture, possible RCM	Shared/cooperative, teleoperation	Piezo, ultrasonic; force: few mN; vel: 0.70 m/s (manual procedure)
Tele-echo, TMS, skin harvesting	Surface tracking	Conventional + dedicated architecture; possible RCM	Autonomous, teleoperation, shared	Conventional; vel: mm/s; force: few N

Table 1: Reference table: surgical domains, tasks, kinematic architectures, control modalities, and sensor/actuator requirements.

- **ROBODOC** (Curexo Technology Corporation): SCARA manipulator plus two final DOF; used for hip and knee surgery; task-level autonomy.
- **MAKO** (Stryker, based on the Acrobot): anthropomorphic manipulator plus three final DOF; used for knee surgery; “hands-on” cooperative or interactive control.

3.2. Workspace Comparison

The SCARA workspace has the shape of a horizontal annular cylinder — a thick disc with a cylindrical void at the centre. When a patient lies supine on the operating table, this workspace envelops the body volume compactly: the intervention volume fits within the reachable workspace with relatively little waste.

The anthropomorphic robot has a workspace shaped like a portion of a sphere. If the patient must lie *inside* this sphere for the robot to reach the surgical site, most of the spherical volume above and to the sides of the patient is wasted. Beyond mere efficiency, a larger workspace has **safety implications**: the workspace is the volume in which, if the robot loses control, it can cause harm. Restricting the operational workspace to only what is needed for the intervention — as the SCARA geometry naturally achieves — is itself a safety measure.

A subtlety worth clarifying. The workspace comparison above concerns the *positional* part only — the locus of reachable wrist-centre positions. The orientation workspace (which poses the wrist can achieve at any given wrist-centre position) is a separate question and is determined by the wrist architecture.

3.3. Other Design Criteria

Beyond workspace shape, two further criteria distinguish the SCARA from the anthropomorphic architecture in the medical context:

Closed-form inverse kinematics. For a medical robot, the ability to solve inverse kinematics analytically (in closed form) rather than numerically is strongly preferred. Closed-form solutions are evaluated by a single algebraic expression at each control cycle: there is no iterative algorithm, no convergence question, and no risk of converging to the wrong solution. The result is faster execution and greater reliability — both critical for patient safety. Both the SCARA and the anthropomorphic architecture admit closed-form inverse kinematics.

Gravitational safety. The SCARA architecture, by design, cannot collapse onto the patient under gravity in the event of a power failure, because gravity acts along its prismatic axis and the arm lies in a horizontal plane. The anthropomorphic robot, whose links are cantilevered, risks collapsing toward the patient if brakes fail or motors lose power. For this reason, anthropomorphic medical robots frequently incorporate counterweights or redundant braking systems.

3.4. Singularities

Singularity. A singularity is a configuration of the robot at which the Jacobian loses rank: the end-effector cannot instantaneously move in one or more directions, and the inverse-velocity problem has no solution (or infinitely many). Singularities can be classified as *arm (structure) singularities*, determined by the positioning chain, and *wrist singularities*, determined by the wrist mechanism.

Arm singularities for both SCARA and anthropomorphic robots can be characterised geometrically and located in the workspace. For example, the classical SCARA elbow singularity occurs when the arm is fully stretched: at the boundary of the reachable workspace. Because the location of these singularities in the workspace is known, the surgeon can position the patient — and the region of surgical interest — well inside the workspace, far from singular configurations. The intervention volume can simply be planned to avoid these regions.

Wrist singularities are different. A standard spherical wrist (three revolute joints whose axes all intersect at a single point) has a singularity whenever two of its axes align. This singularity is not confined to a boundary: it can occur *anywhere* within the manipulator's reachable workspace. Repositioning the patient cannot eliminate it. The solution, as we will see in the following sections, is to design task-specific wrist architectures that avoid this pathology within the relevant configuration space.

4. Case Study: The SCALPP Skin-Harvesting Robot

4.1. Clinical Background

The SCALPP project (Système de Coupe Automatisé pour le Prélèvement de Peau — Automated System for Skin Harvesting) was a European research project that produced a robot for the automatic harvesting of skin grafts. Skin harvesting is needed when tissue damaged by burns, trauma, or cancer must be reconstructed using healthy skin from another body region. The operation is performed with a surgical tool called a *dermatome*, which must slide along the skin surface at a controlled depth and interaction force: only with constant force is the thickness of the harvested graft uniform and clinically usable.

The rationale for robotic assistance is accuracy: a robot can maintain the prescribed contact force and sliding direction with consistency that exceeds the manual performance of even experienced

surgeons.

4.2. Design Constraints

The SCALPP design process began from a set of explicit constraints, which illustrate how clinical requirements translate into kinematic choices:

- For asepsis, non-medical equipment (the robot base, electronics cabinet, and mobile platform) must remain at least 400 mm from the operating table.
- The harvesting zone may range from a few millimetres to 400 mm in length (the length of a thigh). The dermatome orientation may need to rotate up to 90° about the axis normal to the sliding surface (for skull harvesting), and a few degrees about the sliding direction.
- The robot must be operable from both sides of the operating table, to maximise coverage of different body regions, which requires symmetrical joint limits.
- The surgical tool handle must be easily accessible to the nurse for attachment and detachment.
- The motion must be free of singularities and joint limits within the intervention volume.

4.3. Task Description and Architecture Selection

The task is described by three axes in the tool frame: the **approach** axis (z , normal to the skin surface), the **sliding** axis (x , direction of dermatome motion), and the **normal** axis (y). The task proceeds in three phases: approach, slide along the surface, and retract.

The SCARA architecture was selected over the anthropomorphic for the reasons already discussed: the cylindrical workspace better matches the elongated intervention volume along the patient's body, the architecture is inherently fail-safe under gravity, and singularities of the arm can be managed through appropriate workspace planning. Specifically, the arm can be configured in “elbow left” or “elbow right”, a choice left to the surgeon based on the planned position of the surgical team, and the volume of intervention is always placed well inside the workspace to avoid the elbow singularity at its boundary.

The length of the arm links is determined by the workspace requirements (the intervention volume for a full-body patient) and by the asepsis constraint (the robot base must be at least 400 mm from the table).

4.4. Kinematic Architecture Details

4.4.1. Denavit–Hartenberg parameters

The SCALPP robot has a SCARA positioning chain plus a three-DOF wrist, giving six DOF in total. The prismatic joint (joint 1) is placed at the *base* of the serial chain rather than at the end. In industrial SCARA robots it is conventionally at the end (to move the tool vertically without disturbing the horizontal arm); here it serves only to adjust the working height of the arm and does not require highly dynamic maneuvers. Placing it at the base means it moves the entire chain that follows, but since high vertical acceleration is not required, this is acceptable.

The three wrist joints (joints 4, 5, 6) are designed as a **non-spherical wrist** with an offset between axes z_5 and z_6 . This offset deliberately breaks the intersection condition of a spherical wrist, thereby eliminating the spherical-wrist singularity within the reachable workspace. The DH parameters are given in Table 2.

i	α_i	a_i	q_i	d_i
1	0	0	r_1	0
2	0	0	ϑ_2	0
3	0	d_3	ϑ_3	0
4	0	d_4	ϑ_4	r_4
5	$\pi/2$	0	$\vartheta_5 + \pi/2$	0
6	$\pi/2$	d_6	ϑ_6	0

Table 2: Denavit–Hartenberg parameters for the SCALPP robot. Joint 1 is prismatic (r_1); joint 4 is prismatic (r_4); joints 2, 3, 4, 5, 6 are revolute (ϑ_i). Each frame has its z_i axis on joint i . Parameters a_i and α_i are the distance and twist angle from z_{i-1} to z_i , measured along and about x_{i-1} . Parameters d_i and q_i are the distance and angle from x_{i-1} to x_i , measured along and about z_i .

4.4.2. Direct kinematics and singularities

Applying the DH convention and composing the link transformation matrices, the overall homogeneous transformation 0T_6 is:

$${}^0T_6 = \begin{bmatrix} -c_{234}s_5c_6 + s_{234}s_6 & c_{234}s_5s_6 + s_{234}c_6 & c_{234}c_5 & d_4c_{23} + d_3c_2 - d_6c_{234}s_5 \\ -c_{234}s_6 - s_{234}s_5c_6 & -c_{234}c_6 + s_{234}s_5s_6 & s_{234}c_5 & d_4s_{23} + d_3s_2 - d_6s_{234}s_5 \\ c_5c_6 & -c_5s_6 & s_5 & r_1 + r_4 + d_6c_5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The determinant of the Jacobian 6J_6 (expressed in frame 6, which simplifies computation considerably) yields:

$$\det {}^6J_6 = d_3 d_4 \sin \theta_3 \cos \theta_5$$

Two singularities arise:

- $\theta_3 = 0$: the classical SCARA elbow singularity, occurring at the boundary of the workspace when the arm is fully extended. This is managed by keeping the intervention volume well inside the workspace.
- $\theta_5 = \pm\pi/2$: this corresponds to the dermatome being vertical (orthogonal to the skin surface). This configuration is outside the task range and therefore not encountered during normal use.

4.4.3. Inverse kinematics

The inverse kinematics problem is: given a desired end-effector pose

$${}^0T_E = \begin{bmatrix} \mathbf{s} & \mathbf{n} & \mathbf{a} & \mathbf{p} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

find the joint angles. Here $\mathbf{s}$ (sliding), $\mathbf{n}$ (normal), $\mathbf{a}$ (approach) are the unit vectors of the tool frame and $\mathbf{p}$ is the tool origin position.

The solution proceeds in three steps:

1. From the third row of 0T_6 , determine θ_5 , θ_6 , and r_1 as functions of s_z, n_z, a_z, p_z , and obtain the intermediate matrices 4T_5 and 5T_6 .

2. The equation ${}^0T_1 {}^1T_2 {}^2T_3 {}^3T_4 = {}^0T_E {}^ET_6 {}^6T_5 {}^5T_4$ describes the direct kinematics of a SCARA with 4 DOF, where the right-hand side depends only on θ_5 and θ_6 (already solved).
3. Due to joint limits, one of the solutions for θ_5 is excluded, leaving exactly two solutions: “elbow right” and “elbow left”. The surgeon selects the preferred configuration based on operating table layout.

The closed-form nature of this solution is a deliberate design outcome, made possible by the choice of SCARA plus non-spherical wrist architecture.

Classical vs. modified DH. The assignment of frames and derivation of direct kinematics of the SCALPP robot is based on the *modified* convention (proximal, Craig) which places z_i along joint i 's axis and tends to position frames “at the base” of each link. It yields identical forward kinematics as the classical method but the individual transformation matrices differ. In medical robotics literature both conventions are used, so it is essential to check which convention a paper adopts before reading a DH table.

5. Dedicated Wrist Architectures in Medical Robots

5.1. Why Dedicated Wrists?

The spherical wrist, standard in industrial manipulators, has singularities that can occur anywhere inside the workspace. A survey of wrist architectures in medical robots reveals that six distinct types are employed: serial spherical wrist (23%), passive wrist articulation (30%), parallel spherical wrist (20%), parallelogram (9%), circular arc (11%), and cable-pulley (7%).

The distribution reflects the diversity of medical tasks and the need to match the wrist architecture to the constraints of each intervention.

5.1.1. Serial wrist

The conventional serial spherical wrist has three revolute joints whose axes intersect at a point. Compact, well-understood, with closed-form kinematics. Singularities occur when two axes align.

5.1.2. Parallelogram wrist

A parallelogram linkage can be used to realise a remote-centre-of-motion (RCM) effect — constraining the tool to rotate about a fixed external point (such as a trocar insertion point) without requiring that a robot joint coincide with that point. The Da Vinci patient-side manipulator is a prominent example.

5.1.3. Cable-pulley wrist

Cables and pulleys transmit actuation along the kinematic chain, decoupling the actuator location from the joint. This is advantageous when reducing distal mass (for dynamics) or when MR/CT compatibility requires actuators to be remote from the imaging field.

5.1.4. Circular arc wrist

A joint whose motion follows a circular arc centred at the tool tip, naturally implementing an RCM. Three rotational DOF (α , β , γ) orient the tool about a fixed centre.

6. Case Study: The OTELO / MELODY Tele-echography Robot

6.1. System Overview

The OTELO robot (mObile Tele-Ecography using an ultra-Light rObo) was designed within a European research project to enable remote ultrasound examination. An expert sonographer located at a hospital centre remotely operates a lightweight robot placed on the patient by a local nurse. The robot holds and orients an ultrasound probe; the sonographer receives the ultrasound images and a video-conference feed and sends back robot control commands via a standard communication network (3G/WLAN, satellite, or terrestrial link). Today the system is commercialised as the MELODY platform.

The motivation is to bring specialist diagnostic capability to isolated regions without specialist staff. The key engineering challenges are: designing a robot light and safe enough to be placed by a non-specialist nurse; achieving the pointing accuracy required for diagnostic echography; tolerating the latency and variability of non-dedicated communication links; and handling the singularities that arise with the spherical wrist required for probe orientation.

6.2. Kinematic Architecture

The OTELO robot has six DOF, classified as PPRRRP (two prismatic, three revolute, one prismatic from base to end-effector):

- **Joints 1–2 (prismatic):** two translational DOF at the base. These provide limited planar motion on the patient's skin surface and are primarily used for coarse positioning by the nurse before the teleoperation session begins.
- **Joints 3 (revolute):** the first rotational DOF; together with joints 4 and 5 it forms a **spherical wrist** (R_1, R_2, R_3). These three joints orient the probe to the desired pointing direction.
- **Joint 6 (prismatic):** regulates the contact force between the probe and the patient's skin to improve the acoustic coupling. In abdominal scans, pressure is often used to move gas-filled loops of bowel out of the way, which would otherwise block the view of deeper organs like the gallbladder or kidneys.

The spherical wrist (R_1, R_2, R_3) admits closed-form inverse kinematics. One of its singularities corresponds to the probe being in a vertical orientation (normal to the skin surface). During teleoperation, when the expert commands a configuration approaching this singularity, the local robot controller prevents the motion. Since there is a human in the loop, the expert can then adjust the command to approach the desired configuration from a different direction, recovering without disruption to the examination.

A second singularity occurs at the frontier of the workspace and is handled through mechanical or software joint limits. If $\alpha > \pi/4$ (where α is a geometric parameter of the wrist), an additional singularity arises corresponding to z_6 becoming parallel to the plane formed by z_1 and z_2 .

Shared control and singularity management. The OTELO case illustrates a general principle: when a singularity cannot be eliminated by design, and when a human is in the control loop, the system can rely on the human's intent to navigate around it. The robot's controller guards against inadvertent entry into singular configurations, but the human retains the flexibility to redirect the motion. This is fundamentally different from an autonomous system, which would require either a singularity-free architecture or an automatic re-planning algorithm.

6.3. Wrist Architecture Distribution in Echographic Robots

A breakdown of wrist architectures specifically for echographic robots shows: serial spherical wrist (44%), circular arc (25%), and parallelogram (31%). These numbers also illustrate a broader point: **the RCM constraint is not exclusive to MIS**. Even in surface-tracking applications such as echography, the need to orient a probe about a fixed skin contact point makes the RCM design pattern applicable. In fact, 74% of the surveyed echographic robots incorporate an RCM mechanism of some form.

7. Satisfying the Remote Centre of Motion Constraint

7.1. The RCM Constraint in Surgery

The remote centre of motion (RCM) is one of the most important kinematic constraints in medical robotics. It arises in any intervention where a tool or probe must rotate and translate through a fixed point in space — typically the trocar insertion site in MIS, or the needle entry point in interventional radiology. The tool can move freely *inside* the body but the insertion point must not experience lateral forces that would damage tissue.

Four main strategies exist to satisfy the RCM constraint: passive joints, mechanical RCM, kinematic control, and force control. These are not mutually exclusive; real systems may combine them.

7.2. Passive Joints

The simplest realisation uses a kinematically *redundant* chain that includes passive (unactuated) joints. The **Zeus** system illustrates this approach. Its architecture is PRRRR, with the first three joints actuated and a passive universal joint. The trocar physically *forces* the passive universal joint to align with the insertion point: the mechanical structure adapts to the constraint automatically, without any control software needing to enforce it.

Advantages: few motors, no initial precision positioning of the insertion point is needed, and the safety-versus-accuracy trade-off favours safety.

7.3. Mechanical RCM

A mechanical RCM implements the constraint structurally, through a linkage whose kinematics guarantee that a specific point on a distal link is always constrained to remain fixed in space. Two common implementations are:

- **Spherical linkage:** the link segments form a spherical mechanism, ensuring the RCM is at the centre of the sphere. Geometrically precise, but requires complex parts.
- **Parallelogram linkage:** a parallelogram four-bar mechanism is combined with a distal link such that the end-effector always passes through the desired fixed point. The Da Vinci patient-side manipulator uses this architecture. It is compact and manufacturable, and the trocar has no influence on arm motion.

For mechanical RCM, the robot arm must be precisely positioned during setup: a dedicated positioning procedure places the RCM at the planned trocar site before the procedure begins.

7.4. Kinematic Control

In kinematic control, the RCM is not realised mechanically but enforced by a control algorithm in real time. The robot is a standard serial manipulator, and the constraint is modelled as an algebraic equation that the joint velocities must satisfy.

7.4.1. Modelling the RCM constraint

Consider a serial manipulator in which link i is the link that crosses the patient surface at the trocar point P_{trocar} . Let $\mathbf{p}_i(\mathbf{q}(t))$ and $\mathbf{p}_{i+1}(\mathbf{q}(t))$ denote the Cartesian coordinates of the i -th and $(i+1)$ -th joint centres, where $\mathbf{q} = (q_1, \dots, q_n)^T$ is the vector of joint variables.

The RCM point P_{RCM} must lie on link i . It is parameterised as:

$$\mathbf{p}_{\text{RCM}}(\mathbf{q}(t)) = \mathbf{p}_i(\mathbf{q}(t)) + \lambda(t)(\mathbf{p}_{i+1}(\mathbf{q}(t)) - \mathbf{p}_i(\mathbf{q}(t)))$$

where $\lambda \in [0, 1]$ parameterises the position along the link ($\lambda = 0$: joint i coincides with RCM; $\lambda = 1$: joint $i+1$ coincides with RCM).

The RCM constraint requires that this point does not move in the Cartesian space:

$$\dot{\mathbf{p}}_{\text{RCM}} = \mathbf{0}$$

Taking the time derivative:

$$\dot{\mathbf{p}}_{\text{RCM}} = \begin{pmatrix} \mathbf{J}_i + \lambda(\mathbf{J}_{i+1} - \mathbf{J}_i) & \mathbf{p}_{i+1} - \mathbf{p}_i \end{pmatrix} \begin{pmatrix} \dot{\mathbf{q}} \\ \dot{\lambda} \end{pmatrix} = \mathbf{J}_{\text{RCM}}(\mathbf{q}, \lambda) \begin{pmatrix} \dot{\mathbf{q}} \\ \dot{\lambda} \end{pmatrix} = \mathbf{0}$$

This is the constraint equation: the joint velocities $\dot{\mathbf{q}}$ and the rate of change of penetration $\dot{\lambda}$ must lie in the null space of $\mathbf{J}_{\text{RCM}}$. The task-space motions of the end-effector are then planned in the complement of this null space.

Interpretation of λ . The parameter λ captures the *penetration depth* of the tool relative to the trocar. When $\lambda = 1$, the end-effector is at the entry point; as λ decreases toward 0, the tool penetrates deeper. In a control framework, λ can be treated as an additional virtual joint with its own joint limits. A joint limit at $\lambda = 0$ prevents the proximal joint from coinciding with the trocar, which would be a degenerate configuration.

This approach is easy to implement on position-controlled robots and opens the path to *multi-purpose* systems: the same hardware can serve different procedures simply by updating the constraint model and the task specification.

7.5. Force Control

Force control enforces the RCM constraint by measuring the forces the trocar exerts on the robot link and using those measurements to command joint motion that reduces those forces to zero. This approach is conceptually similar to the passive-joint method — the trocar “tells” the robot where the insertion point is — but uses software and sensors instead of mechanical compliance. Force control adds complexity (a force/torque sensor is required) but, like kinematic control, enables multi-purpose systems.

7.6. Comparison of Approaches

8. Kinematic Design and Required Degrees of Freedom

8.1. Task-Driven DOF Analysis

Deciding how many degrees of freedom a medical robot needs requires tracing the clinical task back to a formal mobility analysis. Two exemplary cases are presented.

Method	Key characteristics
Passive joints	Few motors; trocar forces mechanical adaptation; no precise initial positioning needed; safety prioritised over accuracy and rigidity.
Mechanical RCM	Few motors and joints; trocar has no influence on arm motion; requires precise pre-operative positioning (positioning device + procedure).
Kinematic control	Requires kinematic model of the RCM constraint + task; easy to implement on position-controlled robots; enables multi-purpose systems.
Force control	Trocar forces feed a control loop via force/torque sensor; somewhat more complex; also enables multi-purpose systems.

Table 3: Comparison of methods to satisfy the MIS kinematic (RCM) constraint.

8.1.1. Needle positioning in interventional radiology (INNOMOTION)

To position a needle at a target inside the body, the robot needs:

- **3 translations:** to determine the skin entry point.
- **2 rotations** about that entry point: to orient the needle toward the target.

Five DOF suffice. The INNOMOTION system (used for MR-guided interventions) implements exactly this architecture, with actuators that are MR-compatible.

8.1.2. Laparoscopic surgery: the DOF counting problem

In laparoscopic surgery, the surgical tool enters the body through a trocar — a small port. The trocar acts as a *virtual joint* with **four DOF**: translation along the tool axis and three rotational DOF are permitted; lateral translation of the tool axis is constrained.

To determine how many DOF the external arm needs to achieve a desired number of internal DOF (degrees of mobility at the tool tip), we use the **Grübler formula** for linkage mobility.

8.2. The Grübler Formula

Grübler (Chebyshev–Grübler–Kutzbach) formula.

$$m = \sum_{i=1}^n d_i - 6(n + 1 - l)$$

where:

- m : degrees of mobility of the linkage (= degrees of freedom of the tool inside the body in the MIS context);
- d_i : DOF of the i -th joint;
- n : number of joints;
- l : number of rigid bodies, *including* the base.

In a closed loop (as arises in MIS, where the robot and the trocar form a loop), the number of

joints equals the number of rigid bodies: $n = l$. The formula simplifies to:

$$m = (\text{total joint DOF}) - 6$$

The trocar, treated as a virtual joint with 4 DOF, enters the sum. If the desired degrees of mobility are m and the virtual trocar joint contributes 4 DOF, then the number of DOF required in the external arm is:

$$m = 4 + x - 6 \implies x = m + 2$$

8.3. Examples

- If $m = 3$ (e.g., two tilts and one insertion, no tool rotation): $x = 3 + 2 = 5$ external DOF required.
- If $m = 4$ (e.g., two tilts, insertion, and tool axial rotation): $x = 4 + 2 = 6$ external DOF required.

9. Summary and Key Conceptual Threads

- **The kinematic architecture must be matched to the clinical task.** Workspace shape, singularity distribution, inverse kinematics solvability, and gravitational safety are all architecture-level design choices with direct clinical consequences.
- **SCARA architecture offers specific advantages for surface-tracking tasks:** compact workspace fitted to the human body, intrinsic gravitational safety, and an arm singularity that is confined to the workspace boundary.
- **Wrist singularities cannot be eliminated by patient positioning.** They require dedicated wrist designs or control strategies that detect and avoid them at runtime.
- **Closed-form inverse kinematics is a medical design requirement,** not merely a computational convenience. It ensures predictable, fast, and reliable computation at each control cycle.
- **The RCM constraint can be satisfied by four methods:** passive joints, mechanical design, kinematic control, or force control. Each trades off mechanical complexity, software complexity, precision, and flexibility differently.
- **Grübler's formula gives a principled method for counting required DOF** from the desired internal degrees of mobility and the DOF of the virtual trocar joint.

Key topics for revision.

1. The distribution of kinematic architectures across medical robotics domains and why serial architectures dominate.
2. The workspace comparison between SCARA and anthropomorphic robots and why workspace shape is also a safety criterion.
3. The distinction between arm singularities (workspace-bounded, avoidable by positioning) and wrist singularities (workspace-pervasive, requiring architectural or control solutions).
4. The SCALPP robot: design constraints, architecture selection rationale, DH parameters, singularity analysis, and inverse kinematics procedure.

5. The OTELO robot: architecture (PPRRRP), spherical wrist, singularity management via shared control, clinical motivation.
6. The four methods to satisfy the RCM constraint, their trade-offs, and which systems exemplify each.
7. Derivation and application of Grübler's formula to the MIS DOF counting problem.
8. Modelling of the RCM constraint as a velocity constraint and the meaning of the parameter λ .

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